Identity Overview

My name is Saravana Perumal R. I am a Data Scientist and AI practitioner based in Kochi, Kerala, India. I can be reached at +91 9895390670, via WhatsApp at +91 7012753018, or by email at saravanaperumalr14@gmail.com. My GitHub profile is github.com/saravanagithub14 and my LinkedIn is linkedin.com/in/saravanaperumal-r-04b368288.

I represent a unique professional archetype: a biologist-turned-AI practitioner. My career transformation began in biomedical research and biotechnology, where I spent over five years conducting advanced research in tissue engineering, 3D bioprinting, nanomaterial biosynthesis, and stem cell biology. I then transitioned into artificial intelligence and data science, completing a comprehensive diploma program and working as a Data Scientist at Edure, an EdTech company specializing in AI and data science training.

My interdisciplinary background allows me to approach data science problems with rigorous scientific methodology, experimental design thinking, and biological systems intuition. I bridge the gap between life sciences and computational intelligence, bringing research rigor to machine learning projects and practical AI deployment skills to biomedical data analysis.

I am fluent in English with professional proficiency, Tamil as a native language, Malayalam as a native language, and basic proficiency in Hindi. I qualified for the National Eligibility Test (NET) for Assistant Professor in Life Sciences conducted by the University Grants Commission of India in 2019, demonstrating national-level expertise in biological sciences and research methodology.

Professional Summary

I bring over five years of combined experience spanning biomedical research, data analysis, teaching, mentorship, and scientific publishing. My professional journey encompasses roles at premier research institutions including Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST), Central Institute of Brackishwater Aquaculture (CIBA), and Indian Institute of Science Education and Research (IISER) Thiruvananthapuram.

Since February 2024, I have been working as a Data Scientist at Edure in Trivandrum, progressing from Junior Data Scientist to Data Scientist. In this role, I apply advanced analytical techniques to interpret data, build statistical models, and support data-driven decision making. I design and deliver educational resources, workshops, and course content to enhance students' understanding of data science concepts and practical applications. I provide hands-on mentorship to students developing skills in data analysis, machine learning, statistical modeling, and career placement preparation.

My core competencies include Python programming with object-oriented design, SQL database querying, statistical analysis including hypothesis testing and regression techniques, machine

learning model development and deployment, deep learning with TensorFlow and PyTorch, generative AI application development, large language model fine-tuning, retrieval-augmented generation systems, and full-stack deployment using Streamlit, Flask, and cloud platforms.

My problem-solving approach combines scientific rigor with practical engineering. I design experiments systematically, validate hypotheses with statistical methods, iterate on model architectures based on performance metrics, and deploy solutions that balance technical sophistication with operational simplicity. I prioritize interpretability alongside accuracy, documentation alongside development, and long-term maintainability alongside rapid prototyping.

Career Experience: Data Scientist at Edure (February 2025 to Present)

I currently serve as a Data Scientist at Edure, an EdTech company in Trivandrum, India, specializing in data science and artificial intelligence training. This is a full-time position that I have held since February 2025.

In this role, I apply advanced analytical techniques to interpret complex datasets, build statistical models, and support data-driven decision making across educational programs. I design experiments, conduct A/B testing, and analyze student performance metrics to optimize learning outcomes. My work involves developing predictive models for student success, clustering analysis for personalized learning pathways, and time series forecasting for enrollment patterns.

A significant portion of my responsibilities involves training development. I design comprehensive educational resources including video content, coding notebooks, case studies, and hands-on workshops. I deliver lectures on topics ranging from foundational statistics and Python programming to advanced machine learning, deep learning architectures, and generative AI application development. I create curriculum materials that balance theoretical understanding with practical implementation skills.

My mentorship responsibilities include providing one-on-one guidance to students developing technical skills in data analysis, machine learning, statistical modeling, and career placement preparation. I conduct code reviews, project evaluations, and mock interviews. I help students build portfolios, prepare for technical interviews, and navigate career transitions into data science roles. I facilitate study groups, organize hackathons, and coordinate industry interaction sessions.

The technical skills I apply daily include Python programming, SQL querying, statistical analysis with hypothesis testing, machine learning with Scikit-Learn, deep learning with TensorFlow and Keras, generative AI development with LangChain and vector databases, data visualization with Matplotlib and Seaborn, and deployment using Streamlit and Flask.

Career Experience: Junior Data Scientist at Edure (February 2024 to February 2025)

I served as a Junior Data Scientist at Edure from February 2024 to February 2025, marking my formal entry into the data science industry. This role provided intensive hands-on experience in applied data science within an educational technology context.

My primary responsibilities included utilizing advanced analytical techniques to interpret data, build statistical models, and support data-driven decision making. I conducted exploratory data analysis on educational datasets, developed predictive models for student outcomes, and created data pipelines for automated reporting. I worked extensively with Python libraries including Pandas, NumPy, Scikit-Learn, and Matplotlib.

I created and delivered educational resources and workshops to enhance students' understanding of data science concepts. This involved developing coding tutorials, preparing dataset challenges, designing project-based learning modules, and conducting live coding sessions. I taught foundational topics including Python programming, statistical analysis, machine learning algorithms, and data visualization techniques.

I supported students in developing practical skills for industry placement. This included mentoring on portfolio projects, conducting technical skill assessments, providing career guidance, and facilitating connections with hiring partners. I helped students transition from theoretical knowledge to practical implementation capabilities.

The skills I developed during this period include advanced Python programming, machine learning model development and evaluation, statistical hypothesis testing, data visualization and storytelling, curriculum design for technical education, and public speaking for technical audiences.

Career Experience: Senior Research Fellow at SCTIMST (July 2019 to November 2023)

I worked as a Senior Research Fellow in Biomedical Technology at Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST) in Kerala, India, from July 2019 to November 2023. SCTIMST is a premier biomedical research institute under the Government of India, recognized for excellence in medical device development and biomedical research.

My primary research focus involved advanced biomedical technology development, specifically 3D bioprinting of skin using cationized gelatin and oxidized alginate. I developed biofabrication techniques to engineer scaffolds mimicking the extracellular matrix for wound healing and skin grafting applications. This work required expertise in polymeric scaffold development, bio-ink preparation, cell culture, rheological characterization, and molecular analysis.

I led and coordinated interdisciplinary research projects involving materials science, cell biology, and clinical application perspectives. I managed project timelines, coordinated with collaborators, maintained laboratory notebooks, ensured regulatory compliance, and prepared progress reports for funding agencies.

I conducted extensive data analysis using statistical methods and software including Python, GraphPad Prism, and Excel. I performed hypothesis testing, regression analysis, ANOVA, and survival curve analysis. I prepared scientific reports, wrote manuscripts for peer-reviewed journals, and presented findings at internal seminars.

My research portfolio included multiple projects: biosynthesized nanosilver from alginate dialdehyde using green chemistry approaches, development of ferulic acid incorporated antimicrobial self-crosslinking hydrogel for wound healing, evaluation of siRNA role in memory formation using C. elegans as a model organism, and characterization of biomaterial-cell interactions using confocal microscopy and flow cytometry.

The technical skills I developed include cell culture techniques including cryopreservation and cytotoxicity assays, flow cytometry for cell viability and apoptosis analysis, confocal microscopy and image analysis using ImageJ and Fiji, molecular biology techniques including PCR and gel electrophoresis, biomaterial characterization using SEM, TEM, FTIR, and UV-Vis spectrophotometry, 3D bioprinting with extrusion-based systems, electrospinning for nanofiber scaffold fabrication, and statistical data analysis with Python and GraphPad Prism.

Career Experience: Young Professional at CIBA (February 2019 to July 2019)

I worked as a Young Professional II in Vaccine Development at Central Institute of Brackishwater Aquaculture (CIBA) in Chennai, India, from February 2019 to July 2019. CIBA is a research institute under the Indian Council of Agricultural Research focused on aquaculture research and development.

I participated in a vaccine development project targeting Nervous Necrosis Virus in sea bass as part of a multidisciplinary team. My responsibilities included conducting PCR-based disease diagnosis, maintaining viral cultures, preparing vaccine formulations, and performing challenge studies on experimental fish populations.

I gained hands-on experience in PCR amplification and gel electrophoresis for pathogen detection, fish handling and sampling techniques, viral culture and propagation methods, and understanding brackish water aquaculture systems and their disease challenges.

This short-term position provided valuable exposure to applied biological research in an agricultural context and strengthened my molecular biology skill set, particularly in diagnostic applications.

Career Experience: Masters Dissertation Researcher at IISER Thiruvananthapuram (August 2019 to February 2020)

I conducted my Masters dissertation research at the Indian Institute of Science Education and Research (IISER) Thiruvananthapuram from August 2019 to February 2020. My research focused on Periostin-Integrin signaling mechanisms in hematopoiesis and stem cell niche regulation.

I investigated how periostin, an extracellular matrix protein, interacts with integrin receptors to regulate hematopoietic stem and progenitor cell behavior in the bone marrow niche. This involved designing and conducting cell culture experiments, performing immunofluorescence staining, analyzing cell-matrix interactions using microscopy, and synthesizing findings into a comprehensive dissertation manuscript.

The project required extensive literature review on stem cell biology, experimental design with appropriate controls, cell culture technique mastery, fluorescence microscopy and image quantification, and scientific writing for academic documentation.

This research experience deepened my understanding of molecular signaling pathways, experimental design principles, and data interpretation in biological systems. The findings contributed to a book chapter publication in Advances in Experimental Medicine and Biology in 2019.

Educational Background

I completed a Diploma in Data Science from Edure in Trivandrum, India, from 2023 to 2024. This intensive program covered Python programming, statistics and probability, machine learning with Scikit-Learn, deep learning with TensorFlow and PyTorch, natural language processing, computer vision, generative AI and large language models, retrieval-augmented generation systems, model deployment and MLOps, and cloud computing with AWS.

I hold a Master of Science in Biotechnology and Bioinformatics from Kerala University of Fisheries and Ocean Studies (KUFOS) in Kochi, India, completed from 2016 to 2018. This program provided advanced training in molecular biology, genetic engineering, bioinformatics algorithms, protein structure analysis, genomics and proteomics, biostatistics, and research methodology.

I earned a Bachelor of Science in Botany and Biotechnology from the University of Kerala in Trivandrum, India, from 2013 to 2016. This undergraduate program established foundational knowledge in plant biology, microbiology, biochemistry, cell biology, genetic engineering, and biotechnology applications.

Certifications and Achievements

I qualified the National Eligibility Test (NET) for Assistant Professor in Life Sciences conducted by the University Grants Commission of India in 2019. This national-level examination tests expertise in biological sciences and research methodology. Qualifying this examination establishes eligibility for lectureship positions in Indian universities and colleges. It requires comprehensive knowledge across botany, zoology, biochemistry, microbiology, biotechnology, and research methodology. This achievement demonstrates my strong foundation in life sciences and my capability for academic instruction.

Technical Expertise: Programming Languages

I am proficient in Python programming with extensive experience in core programming concepts, object-oriented programming, file handling, and GUI development using Tkinter. I write clean, modular, and well-documented code following PEP 8 style guidelines. I implement classes and objects for complex data structures, use inheritance and polymorphism for code reusability, and design APIs for reproducible workflows.

I am skilled in SQL for database querying and management. I write complex SELECT statements with joins, subqueries, and aggregate functions. I design normalized database schemas, create indexes for query optimization, and implement natural language to SQL translation systems using large language models and LangChain SQL toolkit.

My programming approach emphasizes readability, maintainability, and efficiency. I use version control with Git, write unit tests for critical functions, and document code with docstrings and inline comments.

Technical Expertise: Statistics and Mathematics

I have strong expertise in descriptive statistics including measures of central tendency such as mean, median, and mode, measures of variability such as standard deviation, variance, and interquartile range, and probability distributions including normal, binomial, Poisson, and exponential distributions.

I am proficient in inferential statistics including hypothesis testing with null and alternative hypotheses, confidence intervals for population parameter estimation, t-tests for comparing means between groups, ANOVA for comparing multiple groups, chi-square tests for categorical data analysis, and p-value interpretation for statistical significance determination.

I understand probability theory fundamentals including Bayes theorem for conditional probability, random variables and expected values, probability density functions, and combinatorics for counting problems.

I apply regression techniques extensively including simple linear regression for relationship modeling, multiple linear regression for multivariate prediction, polynomial regression for non-linear relationships, and logistic regression for binary classification problems.

I design experiments using A/B testing methodology, stratified sampling for population representation, random sampling to reduce bias, and control group establishment for causal inference.

Technical Expertise: Data Analysis and Visualization

I conduct comprehensive exploratory data analysis by examining data distributions with histograms and box plots, identifying outliers using statistical methods, analyzing correlations between variables, and discovering patterns through visualization.

I perform data wrangling and cleaning by handling missing values with imputation or removal strategies, removing duplicates to ensure data integrity, correcting data types for proper analysis, and normalizing data for consistency.

I am highly skilled with NumPy for numerical computing including array operations, broadcasting, linear algebra operations, and random number generation. I use Pandas extensively for data manipulation including DataFrame operations, groupby aggregations, merge and join operations, pivot tables for data summarization, time series data handling, and applying custom functions to columns.

I create compelling visualizations using Matplotlib for line plots, scatter plots, histograms, and bar charts with customization options, and Seaborn for statistical visualizations including distribution plots, categorical plots, heatmaps for correlation matrices, and pair plots for multivariate analysis.

I analyze time series data by identifying trends and seasonality, performing moving averages and exponential smoothing, detecting anomalies, and forecasting future values using ARIMA models.

I conduct natural language processing using NLTK for tokenization, stemming, lemmatization, part-of-speech tagging, and named entity recognition, and SpaCy for efficient text processing, dependency parsing, and pre-trained language models.

I perform image processing using OpenCV for image reading and manipulation, filtering and edge detection, object detection, and video processing, and Pillow for basic image operations and format conversions.

I leverage Microsoft Excel for quick data analysis, pivot tables, conditional formatting, and business reporting.

Technical Expertise: Machine Learning

I am proficient with Scikit-Learn for end-to-end machine learning workflows. I implement classification algorithms including Support Vector Machines for margin-based classification, Decision Trees for interpretable rule-based models, Random Forests for ensemble learning with bagging, K-Nearest Neighbors for instance-based learning, Naive Bayes for probabilistic classification, and Logistic Regression for binary and multiclass problems.

I implement regression algorithms including Linear Regression for continuous variable prediction and regularization techniques like Ridge and Lasso for preventing overfitting.

I apply clustering algorithms including K-Means for centroid-based clustering, hierarchical clustering for dendrogram-based grouping, and DBSCAN for density-based clustering.

I perform dimensionality reduction using Principal Component Analysis for feature extraction and variance maximization, and t-SNE for visualization of high-dimensional data.

I conduct feature engineering by creating interaction features, polynomial features for capturing non-linearity, binning continuous variables, and extracting date-time features. I encode categorical variables using one-hot encoding, label encoding, and target encoding. I scale features using standardization and normalization. I select important features using correlation analysis, recursive feature elimination, and feature importance from tree-based models.

I tune hyperparameters using grid search for exhaustive parameter combination testing, random search for efficient exploration, and Bayesian optimization for intelligent search. I validate models using k-fold cross-validation for robust performance estimation, stratified k-fold for imbalanced datasets, and hold-out validation for final model testing.

I deploy machine learning models using Streamlit for interactive web applications, Flask for REST API development, and Tkinter for desktop GUI applications. I track experiments and models using MLflow for version control and reproducibility. I implement CI/CD pipelines for automated testing and deployment. I save models using Pickle and Joblib for serialization.

I have basic experience with AWS services including SageMaker for model training and deployment, S3 for data storage, and Lambda for serverless functions. I containerize applications using Docker for consistent deployment environments.

Technical Expertise: Deep Learning

I am proficient with TensorFlow and Keras for building deep learning models. I construct artificial neural networks with multiple dense layers, activation functions including ReLU, sigmoid, and softmax, and optimization algorithms including Adam, SGD, and RMSprop. I implement backpropagation and gradient descent for training.

I build convolutional neural networks for image classification by designing convolutional layers for feature extraction, pooling layers for spatial dimension reduction, and fully connected layers for classification. I implement architectures including LeNet, AlexNet, VGG, ResNet, and Inception. I apply data augmentation techniques including rotation, flipping, zooming, and shifting to improve model generalization.

I develop recurrent neural networks for sequential data processing including LSTM networks for long-term dependency capture, GRU networks for efficient sequence modeling, and bidirectional RNNs for context from both directions. I apply these to time series forecasting, natural language processing, and speech recognition tasks.

I leverage transfer learning by using pre-trained models including VGG16, ResNet50, InceptionV3, and BERT for fine-tuning on custom datasets. This approach significantly reduces training time and improves performance on limited data.

I implement dropout regularization to prevent overfitting by randomly deactivating neurons during training, batch normalization for stable training, early stopping based on validation performance, and learning rate scheduling for adaptive optimization.

I have foundational knowledge of PyTorch for tensor operations, autograd for automatic differentiation, and building custom neural network architectures using torch.nn module.

Technical Expertise: Generative AI and Large Language Models

I work extensively with large language models including OpenAI GPT models accessed via API, LLaMA3 and LLaMA2 for open-source language generation, and Hugging Face Transformers for accessing and fine-tuning pre-trained models. I understand transformer architecture fundamentals including self-attention mechanisms, multi-head attention, positional encoding, and encoder-decoder structures.

I am highly skilled with LangChain framework for building LLM-powered applications. I construct chains for sequential task execution including simple chains, sequential chains, and router chains. I implement agents for dynamic tool selection and reasoning including ReAct agents, conversational agents, and custom tool-using agents. I configure retrievers for document retrieval including vector store retrievers, contextual compression retrievers, and ensemble retrievers. I manage conversation memory using buffer memory, summary memory, and conversation knowledge graphs. I integrate tools for extended capabilities including search tools, calculator tools, SQL database tools, and custom API tools.

I work with vector databases for semantic search including FAISS for efficient similarity search, ChromaDB for persistent storage, Pinecone for cloud-based vector search, and AstraDB for scalable production deployments. I generate embeddings using OpenAI embeddings, Hugging Face sentence transformers, and custom fine-tuned models.

I implement retrieval-augmented generation systems by chunking documents with optimal size and overlap, embedding chunks into vector space, retrieving relevant context based on query similarity, and generating responses with retrieved context. I optimize RAG systems using hybrid search combining semantic and keyword-based retrieval, reranking retrieved documents for relevance, and query expansion for improved recall.

I perform text summarization using different strategies including stuff chain for short documents, map-reduce chain for large documents, and refine chain for iterative improvement. I implement both extractive summarization selecting important sentences and abstractive summarization generating new summary text.

I fine-tune LLM models using parameter-efficient techniques including LoRA for low-rank adaptation, QLoRA for quantized low-rank adaptation, and full fine-tuning for maximum customization. I use platforms including Lamini AI Cloud for supervised fine-tuning, Hugging Face PEFT library for parameter-efficient methods, and Google Gemma models for domain adaptation.

I deploy generative AI applications using Streamlit for rapid prototyping and deployment, Hugging Face Spaces for cloud hosting, and Gradio for interactive interfaces. I integrate APIs including OpenAI API for GPT models, Ollama for local model serving, GROQ API for high-speed inference on LPUs, and AWS Bedrock for enterprise-grade LLM deployment.

I practice effective prompt engineering by designing clear instructions, providing few-shot examples, implementing chain-of-thought prompting, using system messages for behavior control, and iterating on prompts based on output quality. I develop conversational AI systems with context management, turn-taking logic, and personality consistency.

I use CodeLlama for code generation tasks including multi-language code synthesis, code completion, bug fixing, and code explanation.

Technical Expertise: Biological and Laboratory Skills

I possess extensive laboratory expertise in microbiology including media preparation using aseptic technique, Gram staining for bacterial identification, bacterial culture maintenance, and sterile technique for contamination prevention.

I am skilled in cell culture techniques including cryopreservation of cell lines using DMSO and liquid nitrogen, flow cytometry for cell sorting and viability analysis, cytotoxicity assays including MTT and Alamar Blue assays, stem cell isolation from bone marrow and adipose tissue, cell differentiation protocols, and mammalian cell line maintenance.

I perform histological analysis including H&E staining for tissue morphology examination, immunohistochemistry for protein localization, confocal microscopy for high-resolution imaging, and quantitative image analysis using ImageJ and Fiji software.

I conduct molecular biology experiments including PCR amplification with primer design, gel electrophoresis for DNA and protein analysis, DNA and RNA extraction from tissues and cells, primer design using online tools and BLAST, sequence alignment using NCBI databases, and cloning procedures.

I have expertise in biomaterial engineering including 3D bioprinting with extrusion-based bioprinters, polymeric scaffold development using natural and synthetic polymers, electrospinning for nanofiber fabrication, hydrogel synthesis with tunable mechanical properties, and bio-ink formulation with cells and growth factors.

I operate analytical instrumentation including gas chromatography for compound separation, high-performance liquid chromatography for biomolecule analysis, scanning electron microscopy for surface morphology, transmission electron microscopy for nanoparticle characterization, and UV-Vis spectrophotometry for concentration determination.

Technical Expertise: Research and Academic Skills

I possess strong research methodology skills including designing controlled experiments, formulating testable hypotheses, selecting appropriate statistical tests, and interpreting results in context of existing literature.

I conduct comprehensive literature surveys using databases including PubMed, Google Scholar, Scopus, and Web of Science. I critically evaluate research papers, identify knowledge gaps, and synthesize findings across multiple studies.

I write scientific manuscripts following journal formatting guidelines, structure papers with clear introduction, methods, results, and discussion sections, create figures and tables for data presentation, and respond to peer review comments constructively.

I have contributed to peer-reviewed publications including book chapters and research articles in scientific journals. I understand the publication process from manuscript preparation through revision to final acceptance.

I design experiments with appropriate sample sizes, controls, replicates, and randomization. I apply design of experiments principles for efficient data collection.

I write grant proposals including background and significance sections, research methodology with timelines, budget justification, and expected outcomes.

I develop curriculum materials for technical courses, structure content with learning objectives, create assessment tools, and update materials based on student feedback.

I provide mentorship to junior researchers and students by reviewing their work, providing constructive feedback, setting achievable goals, and supporting their professional development.

I facilitate workshops on technical topics with hands-on coding exercises, live demonstrations, Q&A sessions, and post-workshop support.

I deliver public presentations with clear visual aids, engaging storytelling, technical accuracy, and adaptation to audience expertise level.

Major Project: Simple GenAI App Using Ollama

I built a lightweight local LLM chatbot using Ollama and LangChain for offline text generation without relying on external APIs. This project demonstrates my ability to implement generative AI solutions that run entirely on local infrastructure, providing privacy and eliminating API costs.

The architecture uses Ollama for local model serving, supporting models like Llama2, Mistral, and CodeLlama. I implemented LangChain for conversation management and prompt templating. The application maintains conversation history, handles context window limitations, and provides streaming responses for better user experience.

The technical purpose is to enable organizations and individuals to run AI chatbots without sending data to external servers. This is particularly valuable for sensitive applications in healthcare, legal, and financial domains where data privacy is paramount.

The skills demonstrated include local LLM deployment with Ollama, LangChain framework for conversation management, prompt engineering for consistent responses, memory management for context retention, and Python application development with clean code structure.

Major Project: End-to-End Q&A Chatbot with OpenAI

I designed a context-aware Q&A chatbot integrating the OpenAI API with Streamlit for real-time web-based conversational responses. This project showcases my ability to build production-grade conversational AI applications with modern cloud-based LLMs.

The architecture uses OpenAI GPT models accessed via API with LangChain for orchestration. I implemented conversation memory to maintain context across multiple turns, prompt templates for consistent response formatting, and Streamlit for an interactive web interface with chat history visualization.

The business purpose is to provide customer support automation, internal knowledge base querying, or educational tutoring with natural language interaction. The system reduces response time, improves consistency, and scales efficiently.

The skills demonstrated include OpenAI API integration with error handling and rate limiting, LangChain chains for sequential processing, conversational memory for context retention, prompt design for role-specific behavior, Streamlit deployment for web applications, and user interface design for conversational systems.

Major Project: Q&A Chatbot Using Ollama Open Source

I recreated a Q&A chatbot using Ollama and open-source models for efficient offline inference and low-latency response generation. This project demonstrates my ability to work with open-source alternatives to commercial LLMs while maintaining high quality outputs.

The architecture leverages Ollama for local serving of open-source models including Llama2, Mistral, and Vicuna. I implemented LangChain for prompt management and conversation flow. The system provides streaming responses, handles multi-turn conversations, and operates completely offline.

The technical purpose is to demonstrate that high-quality conversational AI can be achieved without proprietary models or cloud dependencies. This approach reduces costs, eliminates vendor lock-in, and ensures data privacy.

The skills demonstrated include open-source LLM deployment, Ollama configuration and optimization, LangChain integration with local models, memory-efficient conversation management, and performance optimization for inference speed.

Major Project: RAG Document Q&A with GROQ API and Llama3

I built a Retrieval-Augmented Generation chatbot integrating GROQ API and Llama3 with high-speed LPU inference for document-based Q&A. This project showcases my expertise in RAG architecture and integration with cutting-edge inference infrastructure.

The architecture uses GROQ Language Processing Units for ultra-fast inference with Llama3 models. I implemented document chunking with semantic splitting, embedding generation using Hugging Face models, vector storage in ChromaDB, retrieval based on cosine similarity, and context-augmented generation using retrieved chunks.

The business purpose is to enable organizations to query large document collections conversationally, extracting insights from technical manuals, research papers, legal documents, or corporate knowledge bases. The GROQ LPUs provide inference speeds 10x faster than traditional GPU infrastructure.

The skills demonstrated include RAG architecture design, GROQ Cloud API integration, Llama3 prompt engineering, ChromaDB vector database management, embedding model selection and optimization, document processing pipeline development, and semantic search implementation.

Major Project: Conversational Q&A Chatbot with PDF Support

I developed an interactive chatbot capable of processing and conversing with PDF documents using LangChain retrievers and vector stores. This project demonstrates my ability to build document-aware conversational systems that ground responses in uploaded content.

The architecture processes PDFs using PyPDF2 or pdfplumber for text extraction, chunks documents with recursive character splitting for optimal context, embeds chunks using OpenAI or Hugging Face embeddings, stores vectors in ChromaDB with metadata, retrieves relevant chunks based on user queries, and generates answers with retrieved context using GPT or local LLMs.

The technical purpose is to enable users to upload PDF documents and ask questions about their content without manual reading. The system is applicable to research paper analysis, contract review, textbook studying, or technical documentation querying.

The skills demonstrated include PDF text extraction and processing, document chunking strategies, embedding generation and storage, vector similarity search, LangChain retriever configuration, conversational memory with document context, and Streamlit file upload handling.

Major Project: Search Engine GenAI App using Tools and Agents

I implemented a semantic search engine using LangChain Tools and Agents for dynamic data querying and contextual information retrieval. This project showcases my understanding of agentic AI systems that autonomously select and use tools based on user queries.

The architecture defines custom tools for web search, calculator operations, database queries, and API calls. I implemented LangChain agents using ReAct framework for reasoning and acting. The agent analyzes user queries, selects appropriate tools, executes tool calls, and synthesizes results into coherent responses.

The business purpose is to create intelligent assistants that can access multiple data sources, perform computations, and answer complex queries requiring multi-step reasoning. This approach goes beyond simple retrieval to enable autonomous problem-solving.

The skills demonstrated include LangChain tool creation and integration, agent design with reasoning capabilities, tool selection logic implementation, multi-tool coordination, contextual search across diverse data sources, and agentic workflow orchestration.

Major Project: Chat with SQL Database using LangChain SQL Toolkit

I created a natural language interface to query SQL databases using LangChain SQL toolkit and agent-based retrieval. This project demonstrates my ability to bridge natural language and structured databases, making data accessible to non-technical users.

The architecture connects to SQL databases including MySQL, PostgreSQL, and SQLite. I implemented LangChain SQL database wrapper for schema introspection, SQL agent for query generation, query validation to prevent harmful operations, natural language to SQL translation using LLM reasoning, and result formatting for human-readable responses.

The business purpose is to democratize data access by allowing business users to query databases conversationally without SQL knowledge. This reduces dependency on data analysts, accelerates decision making, and improves data literacy.

The skills demonstrated include SQL database connectivity and querying, LangChain SQL toolkit configuration, natural language understanding for query intent, SQL generation with LLMs, query safety validation, agent-based reasoning for complex queries, and Streamlit interface development for database interaction.

Major Project: Text Summarization using LangChain

I developed a text summarization system implementing Stuff, MapReduce, and Refine chains for concise summarization. This project demonstrates my understanding of different summarization strategies and their appropriate use cases.

The architecture implements three LangChain summarization approaches. Stuff chain concatenates entire document into prompt for short texts. MapReduce chain summarizes chunks independently then combines summaries for long documents. Refine chain iteratively improves summary across chunks for highest quality. I allow users to select strategy based on document length and quality requirements.

The technical purpose is to enable efficient information extraction from lengthy documents, reducing reading time while preserving key information. Applications include news article summarization, research paper abstracts, meeting notes condensation, and legal document summaries.

The skills demonstrated include LangChain chain implementation, prompt engineering for summarization quality, document chunking strategies, abstractive summarization with LLMs, summarization quality evaluation, and strategy selection logic based on document characteristics.

Major Project: YouTube and Website Content Summarization

I built an application that extracts and summarizes content from YouTube transcripts and website URLs using LangChain loaders. This project showcases my ability to process diverse content formats and deliver structured summaries.

The architecture extracts YouTube transcripts using YouTube Transcript API, scrapes website content using BeautifulSoup and LangChain document loaders, cleans and preprocesses text, chunks content for processing, and generates summaries using LangChain summarization chains with customizable length and detail level.

The business purpose is to enable rapid content consumption from videos and web pages without watching or reading entirely. This is valuable for research, competitive intelligence, trend monitoring, and educational content curation.

The skills demonstrated include YouTube Transcript API integration, web scraping with BeautifulSoup, LangChain document loaders for diverse formats, text preprocessing and cleaning, chunking strategies for long content, summarization with controllable abstraction level, and multi-format content handling.

Major Project: PDF Query RAG with AstraDB

I built a scalable document query system using AstraDB as a vector store for RAG on PDF documents. This project demonstrates my ability to implement production-grade RAG systems with cloud-based vector databases.

The architecture extracts text from PDFs, chunks documents with semantic splitting, generates embeddings using OpenAI or Hugging Face models, stores vectors in AstraDB with metadata, retrieves relevant chunks based on query similarity, and generates context-grounded responses using LLMs.

The technical purpose is to provide enterprise-scale document querying with high availability, scalability, and performance. AstraDB offers managed vector search, automatic indexing, and integration with Cassandra for hybrid workloads.

The skills demonstrated include AstraDB vector database setup and configuration, PDF processing at scale, embedding generation and batch uploading, vector search optimization, metadata filtering for refined retrieval, RAG pipeline implementation with cloud infrastructure, and production deployment considerations for reliability and performance.

Major Project: Multi-Language Code Assistant using CodeLlama

I developed a coding assistant generating syntax-correct code in multiple programming languages using CodeLlama. This project showcases my ability to leverage specialized LLMs for domain-specific tasks and build developer productivity tools.

The architecture uses CodeLlama models served via Ollama or Hugging Face, LangChain for prompt management and conversation flow, programming language detection from user queries, context-aware code generation with comments and documentation, syntax validation and error checking, and Streamlit interface with code highlighting and copy functionality.

The business purpose is to accelerate software development by automating boilerplate code generation, providing implementation examples, converting code between languages, and explaining complex algorithms. This reduces development time and improves code quality.

The skills demonstrated include CodeLlama model deployment and optimization, prompt engineering for code generation, multi-language code synthesis, syntax validation, developer tool design, code formatting and highlighting, and conversational interface for iterative code refinement.

Major Project: Hybrid Search RAG with Pinecone

I implemented a hybrid retrieval system combining semantic and keyword-based search using Pinecone Vector DB and RRF Fusion. This project demonstrates advanced RAG techniques that improve retrieval quality by leveraging multiple search paradigms.

The architecture performs semantic search using vector embeddings and cosine similarity, keyword search using BM25 algorithm on text content, combines results using Reciprocal Rank Fusion for optimal ranking, and stores vectors in Pinecone with metadata for filtering. I implemented query expansion for improved recall and reranking for precision.

The technical purpose is to overcome limitations of pure semantic search, which may miss exact keyword matches, and pure keyword search, which may miss semantically similar content. Hybrid search provides robustness across query types.

The skills demonstrated include Pinecone vector database configuration, hybrid search algorithm implementation, RRF fusion for result merging, BM25 keyword scoring, semantic embedding generation, query expansion techniques, reranking strategies, and search quality evaluation metrics.

Major Project: Hugging Face Integration Project

I integrated Hugging Face transformer models within LangChain for custom text generation and summarization pipelines. This project demonstrates my ability to work with open-source model ecosystems and build flexible AI pipelines.

The architecture loads models from Hugging Face Hub including BERT, GPT-2, T5, and BART, implements LangChain pipelines for model orchestration, configures generation parameters for temperature, top-p, and max length, implements custom prompts for specific tasks, and deploys models with Hugging Face Spaces or local inference.

The technical purpose is to leverage the extensive Hugging Face model repository for specialized tasks, fine-tune models on custom datasets, and avoid vendor lock-in with commercial APIs. This approach provides flexibility and cost efficiency.

The skills demonstrated include Hugging Face Transformers library usage, model loading and inference, LangChain pipeline configuration, generation parameter optimization, custom tokenization strategies, model fine-tuning preparation, and deployment on Hugging Face infrastructure.

Major Project: Generative AI with AWS Bedrock and Lambda

I deployed GenAI applications on AWS, including Document Q&A with Bedrock and Blog Generation via AWS Lambda. This project showcases my cloud engineering skills and ability to build serverless generative AI solutions.

The architecture uses AWS Bedrock for managed LLM access including Claude, Titan, and Llama2 models, AWS Lambda for serverless function execution, S3 for document storage, API Gateway for REST API exposure, and CloudWatch for logging and monitoring. I implemented RAG pipelines with vector storage in OpenSearch and automated blog generation triggered by Lambda events.

The business purpose is to build scalable, cost-effective generative AI applications without managing infrastructure. Serverless architecture provides automatic scaling, pay-per-use pricing, and high availability.

The skills demonstrated include AWS Bedrock API integration, Lambda function development in Python, S3 storage management, API Gateway configuration, IAM role and policy management, serverless architecture design, event-driven workflows, and cloud cost optimization.

Major Project: Fine-Tuning LLM Models using Lamini and Google Gemma

I executed end-to-end fine-tuning of LLMs using Lamini AI Cloud and Google Gemma with LoRA and QLoRA optimization. This project demonstrates my expertise in adapting pre-trained models to specialized domains and tasks.

The architecture prepares custom datasets in instruction-response format, applies tokenization and preprocessing, implements LoRA for parameter-efficient fine-tuning with low-rank decomposition, applies QLoRA for 4-bit quantization to reduce memory requirements, trains models using Lamini platform with distributed training, and evaluates fine-tuned models on validation sets with perplexity and task-specific metrics.

The technical purpose is to adapt general-purpose LLMs to specific domains, improve performance on specialized tasks, and reduce computational requirements compared to full fine-tuning. This makes custom LLM development accessible.

The skills demonstrated include dataset preparation for fine-tuning, LoRA and QLoRA implementation, Lamini AI Cloud usage, Google Gemma model adaptation, hyperparameter tuning for training, distributed training coordination, model evaluation methodologies, and deployment of fine-tuned models.

Major Project: Real-Time Object Detection in Construction Sites - Constructing Insight

I developed a real-time object detection system for construction entities including buildings, equipment, and workers using the Pictor-v2 dataset and YOLO models. This project demonstrates my computer vision expertise and ability to solve practical industry problems.

The architecture implements YOLO object detection with versions including YOLOv5, YOLOv8, and custom-trained variants. I annotated construction site images with bounding boxes, trained models on GPU infrastructure, optimized inference for real-time performance, and deployed the system with video stream processing.

The business purpose is to improve construction site safety by automatically detecting safety violations, monitor equipment utilization, track worker presence in hazardous zones, and generate automated site activity reports. This reduces accidents and improves operational efficiency.

The skills demonstrated include deep learning for object detection, YOLO architecture implementation, image annotation and dataset preparation, model training and evaluation, real-time video processing, inference optimization, deployment on edge devices, and computer vision for industrial applications.

Major Project: Automated Brain Tumor Detection and Classification Using Deep Learning

I built a CNN-based system for automated brain tumor detection from MRI data, enhanced with Transfer Learning for improved diagnostic accuracy. This project combines my biomedical background with deep learning expertise to address healthcare challenges.

The architecture uses convolutional neural networks with architectures including custom CNNs, VGG16, ResNet50, and InceptionV3. I preprocessed MRI images with normalization and augmentation, implemented transfer learning with pre-trained ImageNet weights, fine-tuned models on tumor datasets, and evaluated performance with accuracy, sensitivity, specificity, and confusion matrices.

The business purpose is to assist radiologists in tumor detection, reduce diagnostic time, improve detection accuracy, and enable early intervention. Automated systems can process large volumes of scans efficiently.

The skills demonstrated include deep learning for medical imaging, CNN architecture design, transfer learning implementation, MRI image preprocessing, data augmentation for limited datasets, model evaluation with clinical metrics, explainable AI with Grad-CAM visualization, and healthcare AI deployment considerations.

Major Project: Tweet Sentiment Sleuth - Decoding Public Opinion with Machine Learning

I analyzed sentiment in 13,870 tweets using an SVM with TF-IDF pipeline, classifying tweets into positive, negative, or neutral categories. This project demonstrates my natural language processing skills and ability to extract insights from social media data.

The architecture preprocesses tweets by removing URLs, mentions, hashtags, and special characters, tokenizes text, removes stopwords, applies lemmatization, extracts TF-IDF features capturing term importance, trains Support Vector Machine classifier with hyperparameter tuning, and evaluates with accuracy, precision, recall, and F1-score.

The business purpose is to understand public opinion on brands, products, or events, monitor brand reputation, analyze customer feedback, detect emerging trends, and inform marketing strategies with data-driven insights.

The skills demonstrated include natural language processing, text preprocessing techniques, TF-IDF feature extraction, Support Vector Machine classification, hyperparameter optimization, sentiment analysis methodology, social media data handling, and insight generation from text data.

Research Project: 3D Bioprinting of Skin Using Cationized Gelatin and Oxidized Alginate

I developed advanced biofabrication techniques at SCTIMST from 2020 to 2023 using cationized gelatin and oxidized alginate to engineer scaffolds mimicking the extracellular matrix for wound healing and skin grafting applications. This project represents my most significant biomedical research contribution.

The work involved chemical modification of gelatin through cationization to enhance cell adhesion, oxidation of alginate to introduce aldehyde groups for crosslinking, bio-ink formulation balancing printability and biocompatibility, 3D bioprinting using extrusion-based systems with optimized parameters, cell encapsulation of fibroblasts and keratinocytes, rheological characterization of bio-inks, mechanical testing of printed scaffolds, in vitro cell viability and proliferation assays, and in vivo wound healing studies in animal models.

The clinical purpose is to provide off-the-shelf skin grafts for burn patients, chronic wound treatment, and surgical reconstruction. Bioprinted skin reduces healing time, minimizes scarring, and improves functional outcomes.

The skills demonstrated include polymeric biomaterial modification, 3D bioprinting technology, bio-ink rheology optimization, cell culture with primary cells, scaffold characterization, wound healing assessment, histological analysis, and translational biomedical research.

Research Project: Biosynthesized Nanosilver from Alginate Dialdehyde

I developed sustainable nanosilver particles at SCTIMST in 2021 using alginate dialdehyde as a natural reducing agent, with applications in biomedical and environmental fields. This project demonstrates my expertise in green nanotechnology.

The work involved oxidizing alginate to produce alginate dialdehyde with reducing properties, synthesizing silver nanoparticles through reduction of silver nitrate, characterizing nanoparticles using UV-Vis spectroscopy for surface plasmon resonance, TEM for size and morphology, XRD for crystallinity, and FTIR for functional groups, testing antimicrobial activity against bacterial pathogens, and evaluating biocompatibility with mammalian cells.

The purpose is to create environmentally friendly nanoparticles avoiding toxic reducing agents, leverage antimicrobial properties for wound dressings, water purification, and food packaging, and demonstrate sustainable nanotechnology approaches.

The skills demonstrated include nanoparticle synthesis, green chemistry principles, nanomaterial characterization techniques, antimicrobial testing, biocompatibility assessment, and sustainable material development.

Research Project: Ferulic Acid Incorporated Anti-Microbial Self-Crosslinking Hydrogel

I developed a self-crosslinking hydrogel at SCTIMST in 2021 incorporating ferulic acid for antimicrobial and wound healing applications. This project showcases my biomaterial engineering capabilities.

The work involved synthesizing hydrogels from natural polymers with self-crosslinking capability, incorporating ferulic acid for antioxidant and antimicrobial properties, characterizing swelling behavior and degradation kinetics, testing mechanical properties for handling, evaluating antimicrobial activity against wound pathogens, assessing biocompatibility and cytotoxicity, and conducting in vitro wound healing assays.

The purpose is to provide wound dressings with inherent antimicrobial properties, reduce infection risk, promote healing through antioxidant effects, and eliminate need for chemical crosslinkers that may be cytotoxic.

The skills demonstrated include hydrogel synthesis and characterization, drug incorporation strategies, antimicrobial testing, mechanical testing, biocompatibility evaluation, and wound healing research.

Research Project: Evaluation of Role of siRNA in Memory Formation in C. elegans

I investigated siRNA-mediated gene silencing in neural plasticity and memory at SCTIMST in 2019 using Caenorhabditis elegans as a model organism. This project demonstrates my molecular biology and neurogenetics expertise.

The work involved designing siRNA sequences targeting memory-related genes, delivering siRNA using feeding or microinjection methods, conducting behavioral assays measuring associative learning, performing quantitative PCR to confirm gene knockdown, analyzing neuronal morphology with fluorescence microscopy, and correlating gene expression changes with behavioral phenotypes.

The purpose is to understand molecular mechanisms of memory formation, identify therapeutic targets for memory disorders, and leverage C. elegans as a tractable genetic model with conserved neural pathways.

The skills demonstrated include RNA interference techniques, model organism research, behavioral genetics, quantitative PCR, neurobiological methods, and functional genomics.

Research Project: Periostin-Integrin Signaling in Hematopoiesis

I studied how periostin-integrin signaling and extracellular matrix components regulate stem and progenitor cell behavior at IISER Thiruvananthapuram in 2019 for my Masters dissertation. This project deepened my stem cell biology knowledge.

The work involved culturing hematopoietic stem and progenitor cells, treating cells with recombinant periostin protein, measuring cell adhesion to ECM proteins, performing

immunofluorescence staining for integrin expression, quantifying cell proliferation and differentiation, analyzing signaling pathway activation with Western blotting, and correlating ECM interactions with stem cell maintenance.

The purpose is to understand stem cell niche regulation, identify factors controlling stem cell fate, and inform regenerative medicine strategies for blood disorders.

The skills demonstrated include stem cell culture, immunofluorescence microscopy, cell adhesion assays, Western blotting, cell signaling analysis, and stem cell biology research.

Publications and Scientific Contributions

I contributed a book chapter titled "Emerging Strategies in Bone Tissue Engineering" published in 2022 in the textbook Tissue Engineering, pages 469 to 492. This chapter reviews cutting-edge approaches in bone regeneration including scaffold design, growth factor delivery, and stem cell-based therapies.

I published a research article titled "Biosynthesized Nanosilver from Alginate Dialdehyde: An In Vitro Evaluation" in SSRN Electronic Journal in 2021. This paper details the green synthesis of silver nanoparticles and their antimicrobial properties.

I published a research article titled "Ferulic Acid Incorporated Anti-Microbial Self-Crosslinking Hydrogel" in SSRN Electronic Journal in 2021. This paper describes the development and characterization of a novel wound dressing material.

I contributed a book chapter titled "Periostin and Integrin Signaling in Stem Cell Regulation" published in 2019 in Advances in Experimental Medicine and Biology. This chapter discusses molecular mechanisms of stem cell niche regulation.

These publications demonstrate my ability to communicate complex scientific findings, contribute to peer-reviewed literature, and advance knowledge in biomedical research fields.

Teaching and Mentorship Philosophy

I believe effective teaching combines theoretical understanding with practical application. I design learning experiences that progress from foundational concepts to hands-on projects, ensuring students develop both knowledge and skills.

My teaching methodology emphasizes active learning through coding exercises, case studies, and project-based assignments. I provide immediate feedback during live coding sessions, encourage students to debug errors independently, and create a safe environment for asking questions.

I structure courses with clear learning objectives, modular content that builds progressively, regular assessments to track understanding, and real-world examples that demonstrate relevance. I adapt teaching pace based on student comprehension and provide additional resources for advanced learners.

My mentorship approach focuses on individual student development. I conduct one-on-one sessions to understand career goals, provide personalized project guidance, review code with constructive feedback, and help students build portfolios. I connect students with industry professionals, share job opportunities, and conduct mock interviews.

I believe mentorship extends beyond technical skills to include communication abilities, problem-solving approaches, professional behavior, and continuous learning habits. I encourage students to contribute to open-source projects, participate in competitions, and build online presence through blogs and GitHub repositories.

I find fulfillment in witnessing student growth, celebrating their achievements, and maintaining long-term relationships beyond formal instruction. Many of my mentees have successfully transitioned into data science roles, and I continue to support their career progression.

Personal Working Style

I approach problems systematically by first understanding requirements thoroughly, breaking complex tasks into manageable components, prototyping solutions quickly, and iterating based on feedback. I prioritize clarity and maintainability in code, believing that readable code is more valuable than clever but obscure implementations.

I value collaboration and communication. I document my work comprehensively, explain technical concepts to non-technical stakeholders, seek feedback regularly, and share knowledge through mentorship and teaching. I believe the best solutions emerge from diverse perspectives.

I practice continuous learning by reading research papers, experimenting with new frameworks, participating in online courses, and contributing to technical communities. The rapid evolution of AI and data science requires constant upskilling, and I dedicate time weekly to learning new techniques.

I maintain work-life balance by setting clear boundaries, practicing effective time management, and engaging in activities outside work. I believe sustainable productivity comes from rest and diverse experiences.

I approach challenges with scientific rigor. I formulate hypotheses about why problems occur, design experiments to test solutions, measure outcomes quantitatively, and document learnings for future reference. This methodology stems from my research background and applies equally to debugging code and optimizing models.

I value ethical AI development. I consider bias in datasets, fairness in model predictions, transparency in decision-making systems, privacy in data handling, and societal impact of deployed systems. I believe technologists have responsibility to build AI that benefits humanity.

Long-Term Vision

I envision myself as a leader in biomedical AI, bridging my biological sciences foundation with advanced artificial intelligence capabilities. I aim to develop AI systems that solve critical healthcare challenges including drug discovery, personalized medicine, medical imaging diagnostics, and clinical decision support.

In the next three to five years, I plan to deepen my expertise in deep learning for medical applications, gain experience working with electronic health records, collaborate with clinical researchers on AI projects, and contribute to open-source medical AI tools.

I aspire to pursue advanced education through a PhD program in biomedical informatics or computational biology, conducting research at the intersection of AI and life sciences. I am particularly interested in multi-modal learning combining imaging, genomics, and clinical data for comprehensive patient analysis.

I want to establish myself as a thought leader by publishing research papers, speaking at conferences, contributing to standards for medical AI, and mentoring next-generation researchers. I believe AI will revolutionize healthcare, and I want to contribute meaningfully to this transformation.

Beyond technical achievements, I aim to build educational programs that make biomedical AI accessible to biologists and medical professionals. I want to create bridges between disciplines, teaching computer scientists about biological complexity and teaching biologists about computational tools.

I envision founding or leading research teams focused on translating AI research into clinical practice. I am passionate about ensuring that AI innovations reach patients, improve outcomes, and increase healthcare access globally.

I remain committed to ethical AI development, advocating for responsible use of AI in healthcare, transparency in algorithmic decision-making, and equitable access to AI-powered medical technologies.

Retrieval Tags

Data scientist, machine learning engineer, AI practitioner, biologist turned data scientist, biomedical research background, Edure data scientist, Trivandrum Kerala India, SCTIMST research fellow, 3D bioprinting skin tissue engineering, Python programming expert, SQL database querying, statistical analysis hypothesis testing, scikit-learn machine learning,

TensorFlow Keras deep learning, PyTorch neural networks, generative AI large language models, LangChain framework, retrieval augmented generation RAG, vector databases FAISS ChromaDB Pinecone AstraDB, OpenAI GPT API integration, Llama3 open source LLM, Hugging Face transformers, prompt engineering, conversational AI chatbot development, LLM fine-tuning LoRA QLoRA, AWS Bedrock cloud deployment, Docker containerization, Streamlit Flask web applications, computer vision YOLO object detection, CNN image classification, natural language processing sentiment analysis, medical imaging brain tumor detection, biotechnology bioinformatics background, cell culture flow cytometry, molecular biology PCR gel electrophoresis, biomaterial engineering hydrogel scaffold development, nanotechnology silver nanoparticle synthesis, stem cell biology hematopoiesis research, scientific publication peer reviewed journals, teaching mentorship curriculum development, NET qualified assistant professor eligibility, data visualization Matplotlib Seaborn, pandas numpy data wrangling, exploratory data analysis EDA, feature engineering hyperparameter tuning, model deployment MLflow CI/CD, time series analysis forecasting, A/B testing experimental design, confidence intervals t-tests ANOVA, logistic regression classification, K-means clustering dimensionality reduction PCA, random forest decision trees, support vector machines SVM, transfer learning pre-trained models, image processing OpenCV Pillow, text summarization document Q&A, YouTube content extraction web scraping, hybrid search semantic keyword retrieval, CodeLlama code generation, Ollama local LLM serving, GROQ API high speed inference, SQL toolkit natural language to database, agentic AI tools and agents, educational technology EdTech, workshop facilitation public speaking, research methodology literature review, grant writing scientific writing, interdisciplinary problem solving, career transition biology to AI, Kerala University KUFOS IISER education, Tamil Malayalam English multilingual, wound healing regenerative medicine, confocal microscopy histology immunohistochemistry, UV-Vis spectrophotometry SEM TEM characterization, electrospinning nanofiber fabrication, vaccine development aquaculture, brackish water CIBA Chennai, periostin integrin signaling pathways, extracellular matrix ECM, C elegans model organism research, siRNA gene silencing memory formation, Bacopa monnieri plant biotechnology, ethical AI development responsible technology, continuous learning professional development, portfolio project demonstration